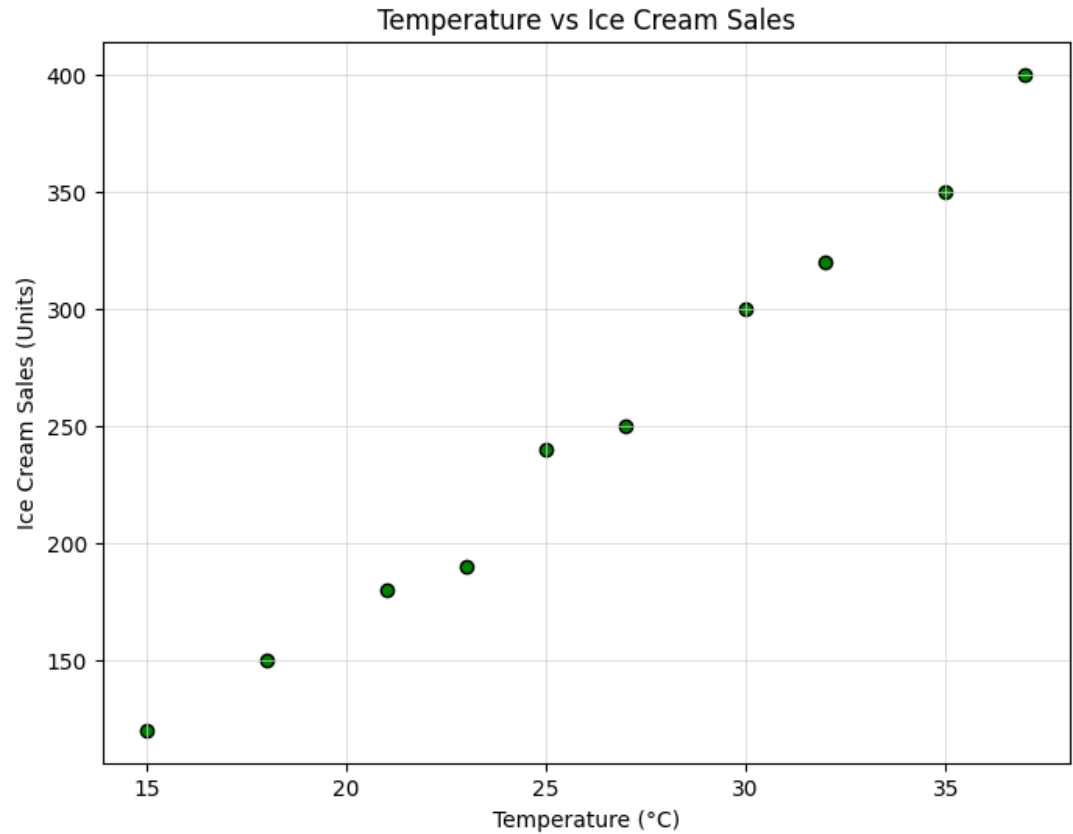


SCATTER PLOT

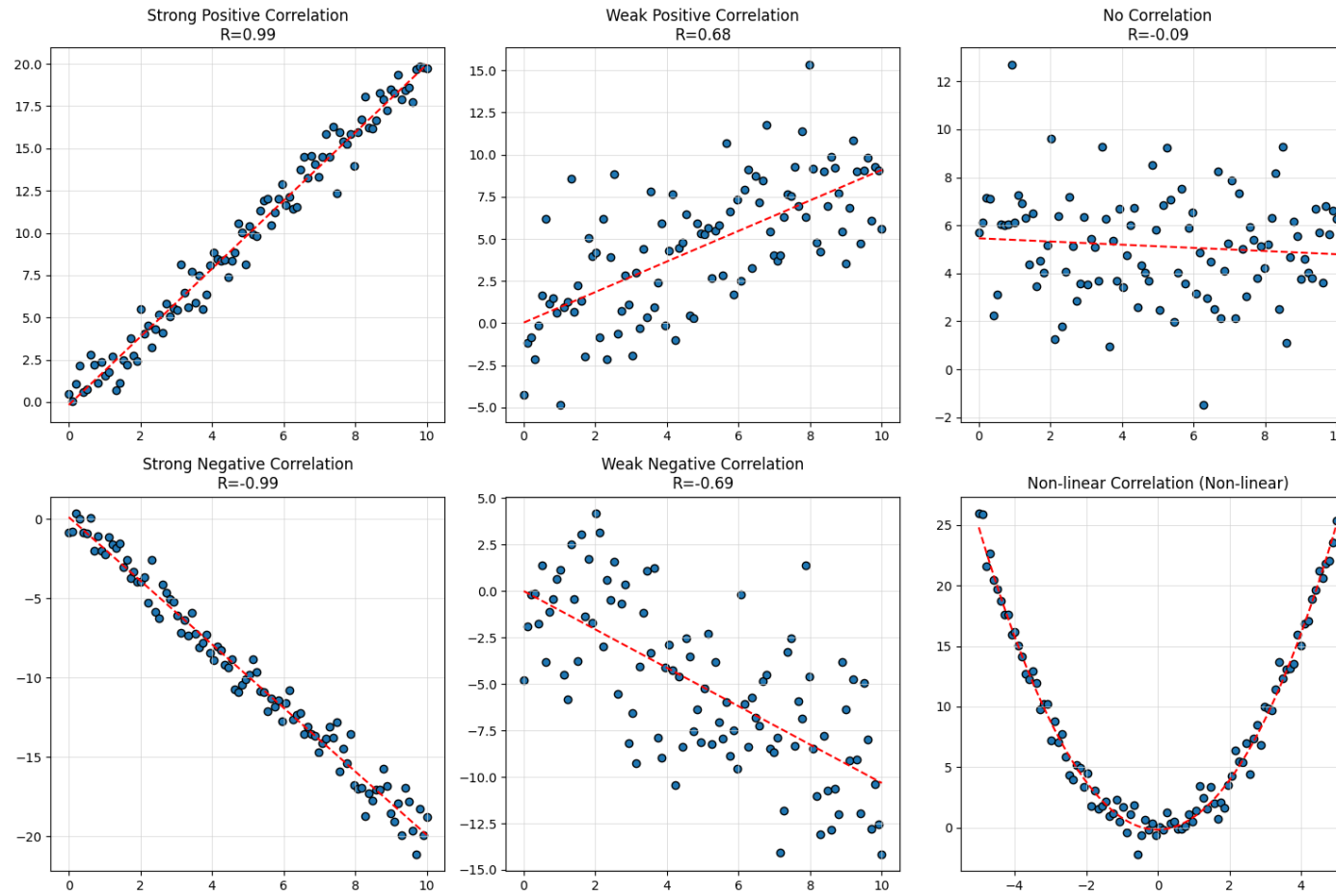
Temperature (°C)	Ice Cream Sales (Units)
15	120
18	150
21	180
23	190
25	240
27	250
30	300
32	320
35	350
37	400



CORRELATION

- ❖ Correlation Coefficient: A numerical value quantifying the strength and direction of the relationship between two variables in a scatter plot.
- ❖ Range:
 - ❖ +1: Perfect positive correlation (as one variable increases, the other increases proportionally).
 - ❖ 0: No correlation (no linear relationship between variables).
 - ❖ -1: Perfect negative correlation (as one variable increases, the other decreases proportionally).

CORRELATION



CORRELATION

- ❖ The formula for calculating Pearson's correlation coefficient (r) is:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

- Where:
 - n = number of data points
 - x and y = individual data points for two variables
 - Σ = summation symbol

- ❖ It analyze linear relationships between variables.

Hours Studied (X)	Test Score % (Y)
20	40
24	55
46	69
62	83
22	27
37	44
45	61
27	33
65	71
23	37

CORRELATION

❖ The formula for calculating Pearson's correlation coefficient (r) is:

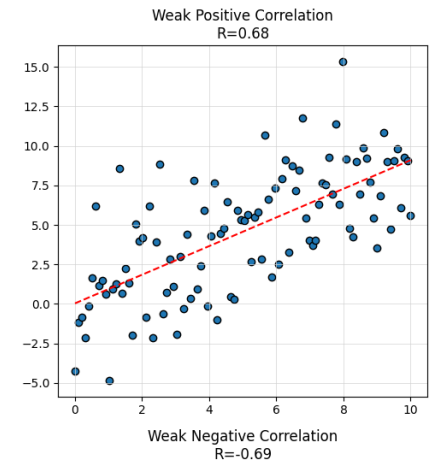
$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

- Where:
 - n = number of data points
 - x and y = individual data points for two variables
 - Σ = summation symbol

	Hours Studied (X)	Test Score % (Y)	XY	X ²	Y ²
	20	40	800	400	1600
	24	55	1320	576	3025
	46	69	3174	2116	4761
	62	83	5146	3844	6889
	22	27	594	484	729
	37	44	1628	1369	1936
	45	61	2745	2025	3721
	27	33	891	729	1089
	65	71	4615	4225	5041
	23	37	851	529	1369
SUM	371	520	21764	16297	30160

$$r = 0.879350768$$

ASSUMPTIONS OF PEARSON CORRELATION



❖ 1. Linearity:

- ❖ The relationship between the two variables should be linear.
- ❖ A scatterplot can be used to visually inspect if a linear trend exists between the variables.

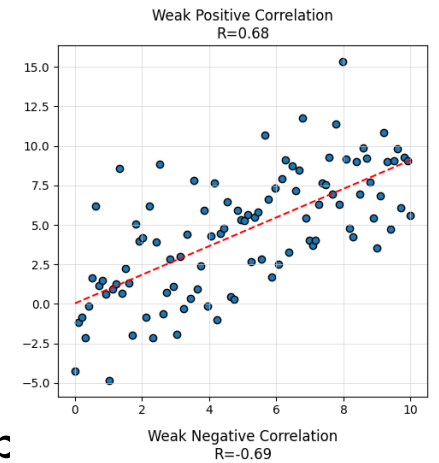
❖ 2. Continuous Data:

- ❖ Both variables must be continuous (interval or ratio scale).
- ❖ Examples include height, weight, and temperature.

❖ 3. Homoscedasticity:

- ❖ The variance of the error terms should be constant across all levels of the independent variable.
- ❖ This means the spread of residuals (errors) should be roughly the same across all values.

ASSUMPTIONS OF PEARSON CORRELATION



❖ 4. Normality:

- ❖ Both variables should follow a normal distribution, especially when conducting hypothesis tests on the correlation.
- ❖ This can be tested using tools like a Q-Q plot, Anderson Darling test or Shapiro-Wilk test.

❖ 5. No Outliers:

- ❖ Outliers can significantly distort the value of the Pearson correlation coefficient.
- ❖ It is important to check for and handle any outliers appropriately, using robust statistical techniques or removing them if justified.

❖ 6. Paired Observations:

- ❖ Each data point in the dataset should consist of a paired observation (i.e., both variables are measured for each case).

CORRELATION DOES NOT IMPLY CAUSATION

❖ Correlation:

- ❖ Indicates that two variables are related, meaning changes in one variable are associated with changes in the other.
- ❖ Example: Ice cream sales and temperature are correlated, as higher temperatures tend to increase ice cream sales.

❖ Correlation Causation:

- ❖ Correlation does not imply causation.
- ❖ There may be a third variable (confounder) influencing both correlated variables.
- ❖ Spurious correlations can occur, where variables appear related but are not causally linked.

❖ Example of Misinterpretation:

- ❖ High chocolate consumption is correlated with more Nobel Prize winners in certain countries, but eating chocolate does not cause someone to win a Nobel Prize.

COEFFICIENT OF DETERMINATION (R^2)

- ❖ Measures the proportion of variance in the dependent variable that is explained by the independent variable(s) in a regression model.
- ❖ Values range from 0 to 1:
 - ❖ $R^2 = 1$: The model explains all the variance in the dependent variable.
 - ❖ $R^2 = 0$: The model explains none of the variance in the dependent variable.
- ❖ In simple linear regression (with only one independent variable), R^2 is the square of the correlation coefficient r .
- ❖ $R^2 = r^2$

COEFFICIENT OF DETERMINATION (R^2)

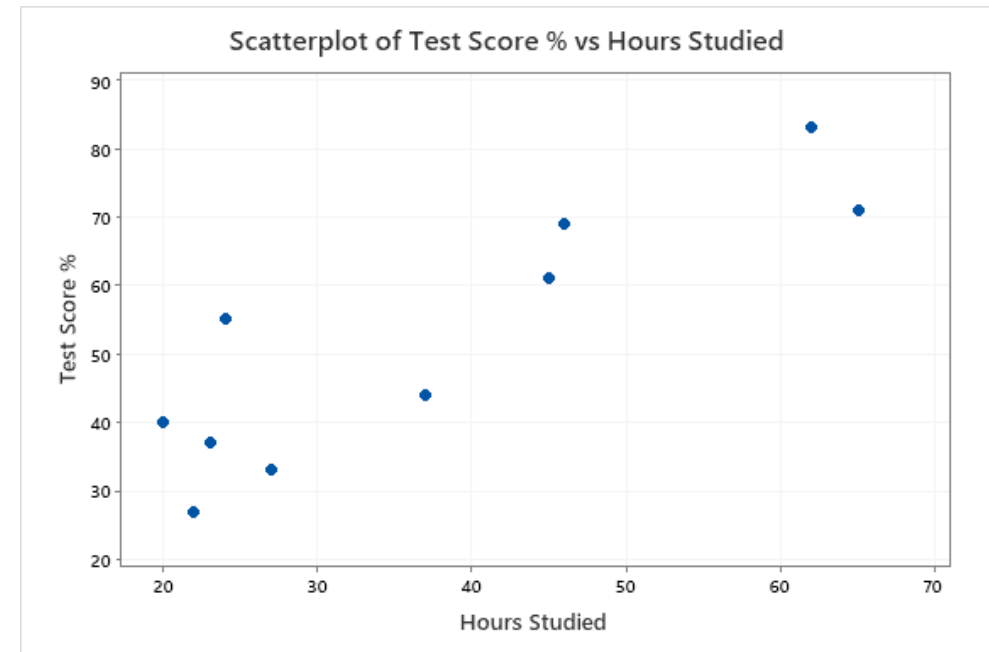
- ❖ For example, if the correlation coefficient r between two variables is 0.8, the coefficient of determination R^2 would be:
- ❖ $R^2 = 0.8^2 = 0.64$
- ❖ This means that 64% of the variance in the dependent variable is explained by the independent variable in the regression model.

SIMPLE LINEAR REGRESSION

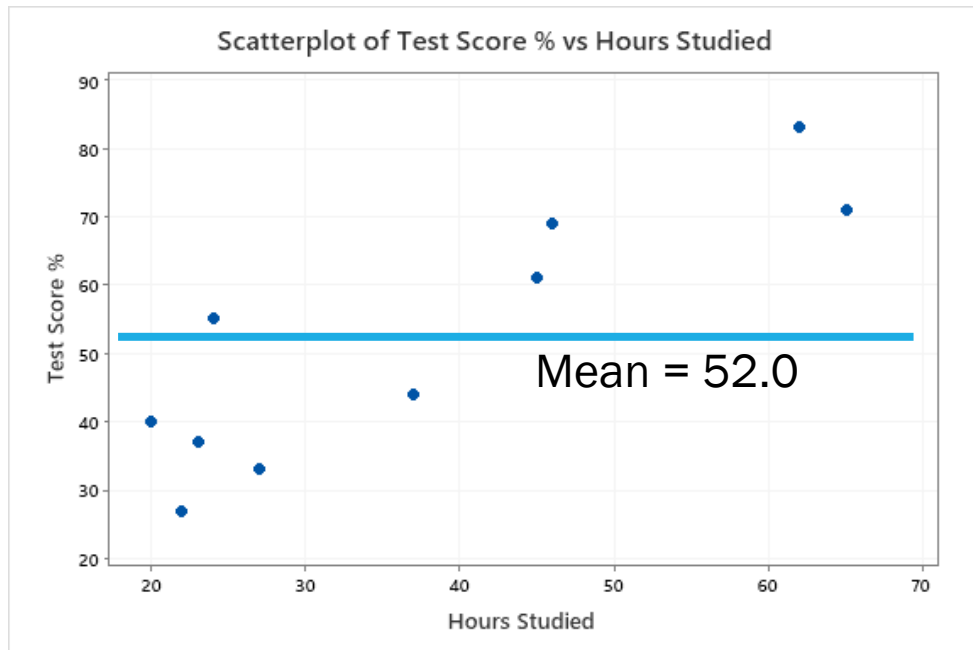
- ❖ Simple linear regression is a statistical method that models the relationship between two variables by fitting a linear equation to observed data.
- ❖ The model explains how the dependent variable (Y) changes when the independent variable (X) changes.
- ❖ The equation for a simple linear regression is:
 - ❖ $Y = \beta_0 + \beta_1 X + \epsilon$
 - ❖ Y = Dependent variable (the outcome we're predicting)
 - ❖ X = Independent variable (the predictor)
 - ❖ β_0 = Intercept (value of Y when X=0)
 - ❖ β_1 = Slope (the change in Y for a one-unit change in X)
 - ❖ ϵ = Error term (the difference between the observed and predicted values)

SIMPLE LINEAR REGRESSION

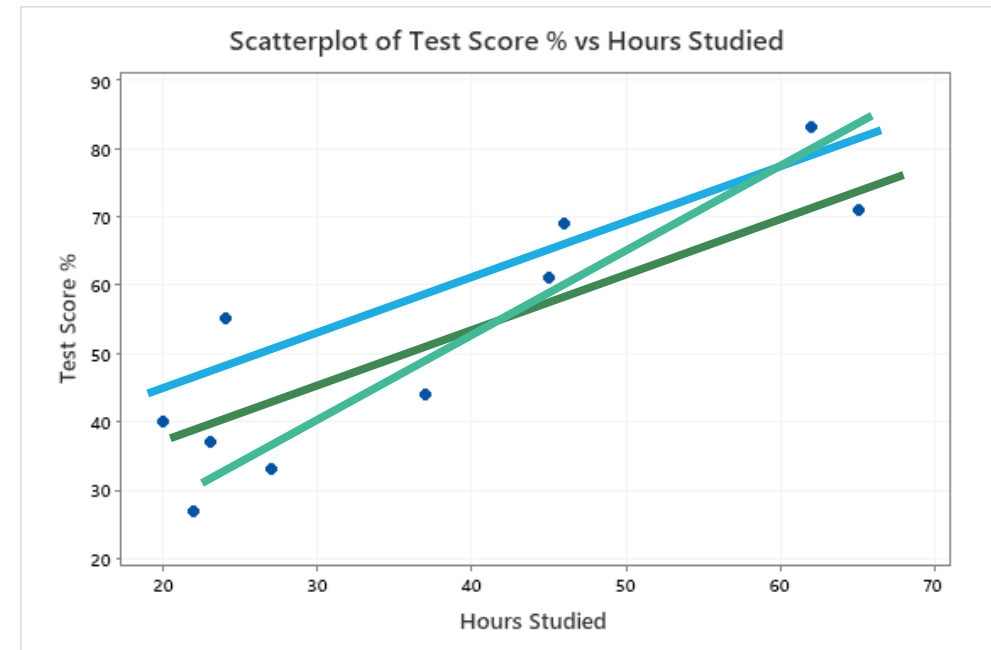
Hours Studied (X)	Test Score % (Y)
20	40
24	55
46	69
62	83
22	27
37	44
45	61
27	33
65	71
23	37



SIMPLE LINEAR REGRESSION



OR



SIMPLE LINEAR REGRESSION

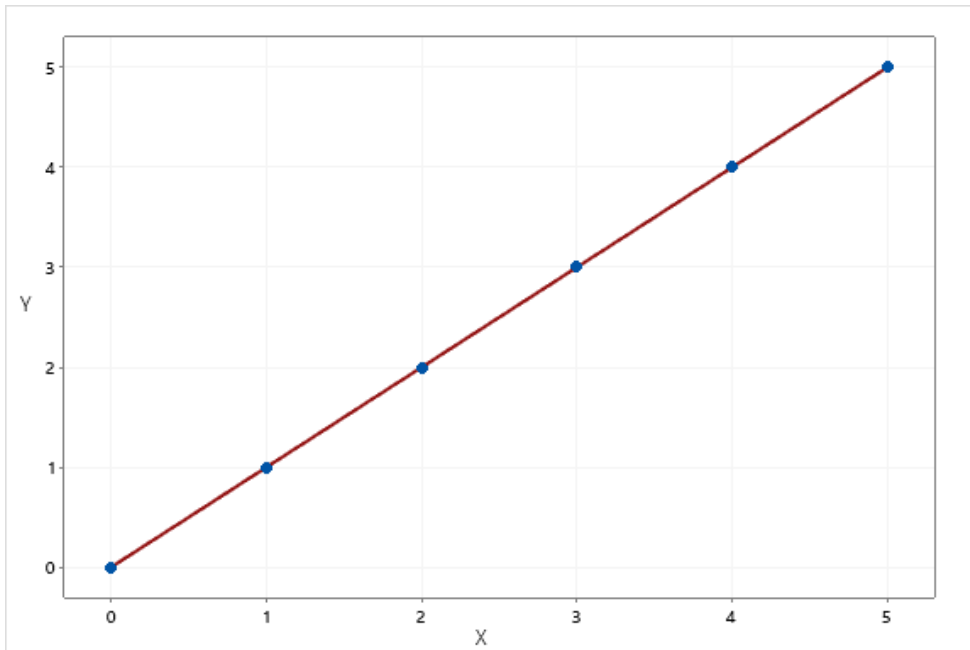
$$Y = b_0 + b_1X$$

Y is the dependent variable

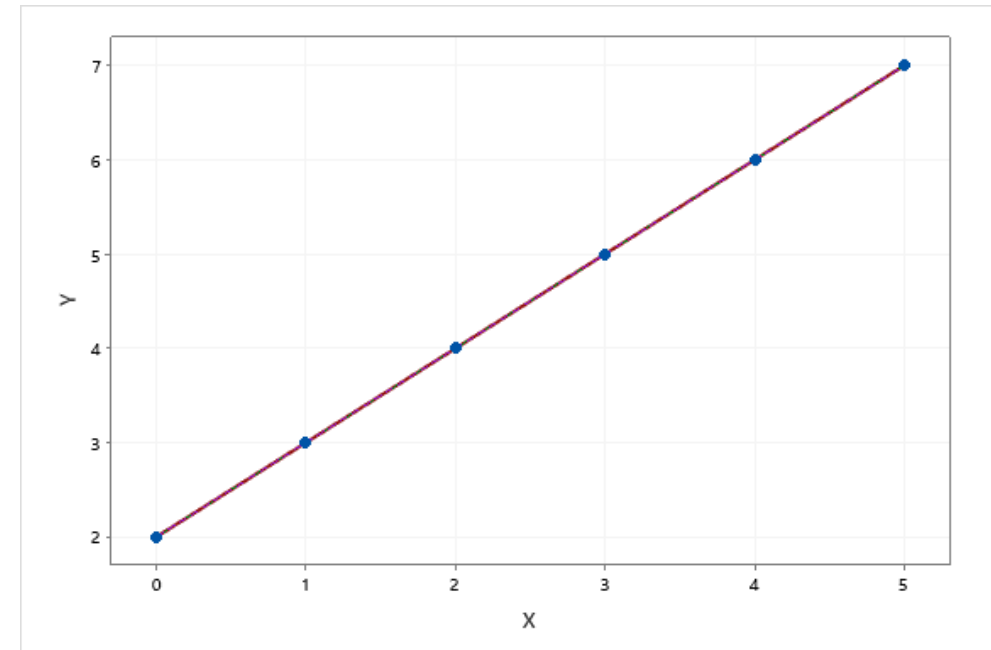
X is the independent variable

b_1 is the slope of the line

b_0 is the y-intercept

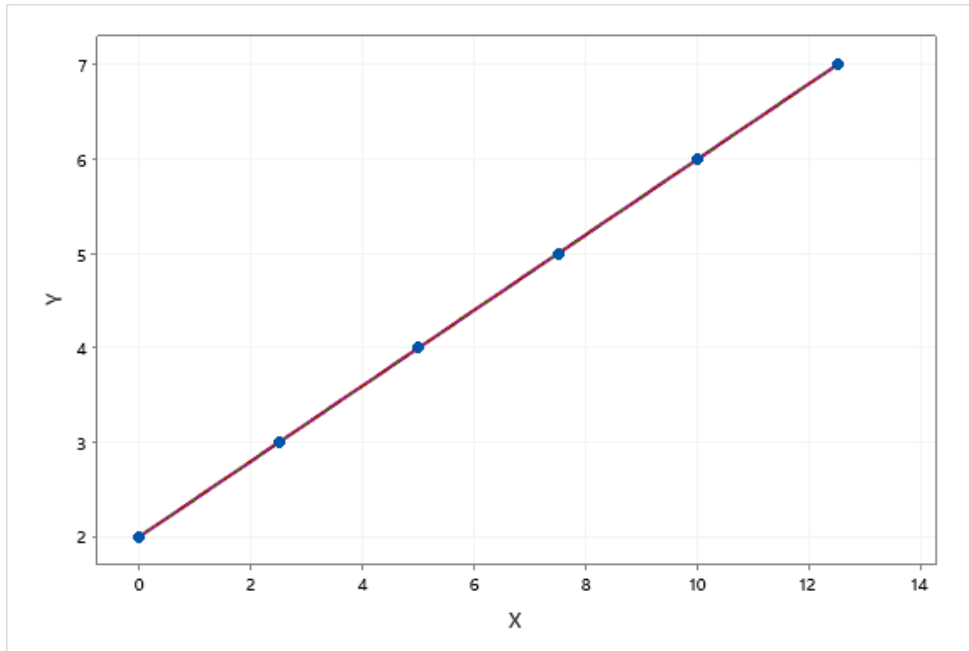


$$Y = 0.00 + 1.00 X$$

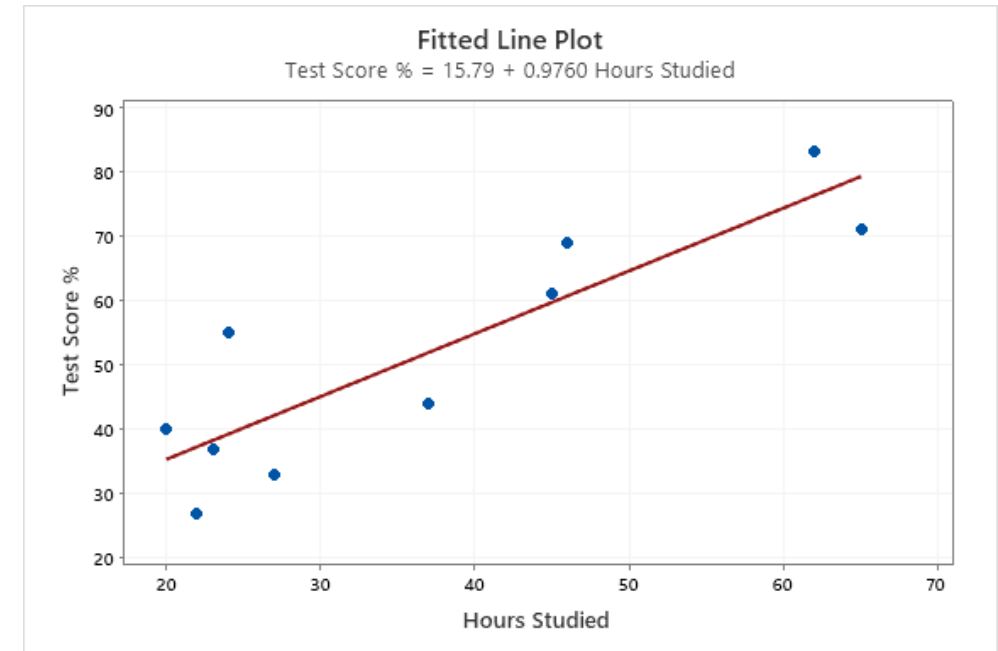


$$Y = 2.00 + 1.00 X$$

SIMPLE LINEAR REGRESSION



$$Y = 2.00 + 0.40 X$$



$$Y = 15.79 + 0.9760 X$$

SIMPLE LINEAR REGRESSION

x	y	x^2	y^2	xy
20	40			
24	55			
46	69			
62	83			
22	27			
37	44			
45	61			
27	33			
65	71			
23	37			
$\Sigma x = 371$	$\Sigma y = 520$			

$$\hat{y} = b_0 + b_1x$$

$$b_1 = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}$$

$$b_0 = \frac{\Sigma y - b_1 \Sigma x}{n}$$

$$\hat{y} = 15.79 + 0.9760 x$$